Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/10/2008
PI Name:	Moore, Steven T. Ph.D.		
Project Title:	Galvanic Vestibular Stimulation (GVS) Augmented Training for Exploration Class Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	s.moore@cqu.edu.au	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	212-241-1943
Organization Name:	Mount Sinai School of Medicine		
PI Address 1:	Human Aerospace Laboratory		
PI Address 2:	Department of Neurology		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	14
Comments:	NOTE: PI moved to Central Queensland University, Australia, July 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2005	**End Date:**	04/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SM00801		
Performance Goal No.:			
Performance Goal Text:			
	1. Original Project Aims Space Adaptation Syndrome (SAS) is a generalized term for various neurovestibular and autonomic symptoms experienced following gravity transitions, such as insertion into orbit and landing on the Earth, moon or mars. Although many of the frank symptoms of motion sickness can be alleviated by prophylactic drugs, the problem remains of disruption to sensorimotor patterns of gait and head movement stabilization. This presents a danger to astronauts during manned docking and landing maneuvers of the spacecraft, and post-landing emergency egress or EVA. To a certain extent SAS can be modeled on Earth following hyper-g centrifugation, but the neurovestibular component is short-lived and usually accompanied by nausea. We have developed a new system for modeling the postural, locomotor and perceptual effects of SAS using galvanic vestibular stimulation (GVS) that does not provoke motion sickness. The primary aim of the current proposal is to develop training protocols that utilize GVS to enhance crew performance on exploration class missions (GVS-augmented training).		

Task Description:	2. Key Findings a) We have completed two studies involving 32 normal subjects that demonstrated that acute GVS exposure accurately modeled postural instability and locomotor and gaze dysfunction observed following spaceflight. GVS was well-tolerated, did not induce symptoms of motion sickness, and the effects were reversible (only lasting as long as the current was applied). These results have been published in Experimental Brain Research. b) A pilot study of the subjective response to GVS was conducted at the NSBRI Integration Facility in Houston on May 1st, 2006. Five veteran astronauts (4 shuttle, 1 ISS) participated in the demonstration of GVS technology, which showed that the system generates not only the physiological effects of gravity transitions and re-adaptation to Earth, but also the illusory sensations of motion experienced post-flight. Subjects were exposed to a pseudorandom stimulus, head-driven, and a combination of the two. All shuttle crewmembers (4) stated that the sensations of motion and instability were remarkably similar to that experienced on landing day. The former ISS crewmember felt that the GVS stimulus was not as strong as landing day, and more akin to R+3. These preliminary observations are encouraging as GVS not only accurately reproduced the postural, locomotor and visual post-flight deficits, but also recreated the subjective sensations experienced by astronauts after spaceflight. Moreover, the amplitude of the GVS current used was proportional to the length of exposure to microgravity in the veteran astronauts. Thus, GVS has the potential to provide a high-fidelity ground-based analog of the physiological and perceptual effects of microgravity exposure that can be adjusted to suit mission duration. c) GVS was applied during simulated orbiter landings in both a commercial flight simulator (Airbus A330, Toulouse, France) and the VMS (NASA Ames). The results demonstrated that GVS induced significant roll oscillation of the aircraft during final approach and yaw oscillations during rollout, increased pilot workload, and decreased visual acuity. 3. Impact of Findings Our initial studies have demonstrated that ambulatory Galvanic Vestibular Stimulation (GVS) system shows significant potential as a high-fidelity simulation of Space Adaptation Syndrome (SAS) and Entry and Landing Syndrome (ELS). Future work will concentrate on developing computer-generated Galvanic current waveforms to model various training scenarios: ELS short-duration (shuttle); ELS long-duration (ISS, exploration class missions); EVA; orbiter landing (VMS) etc 4. Proposed work for the coming year a) Conduct formal study of perceptual response to GVS in veteran astronauts. b) Further develop GVS patterns to simulate missions of various durations. c) Conduct formal study of effects of GVS on pilot performance during simulated orbiter landings in the VMS, and compare to data from actual landings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Galvanic Vestibular Stimulation (GVS) system developed to model the effects of spaceflight has also proven to model vestibular pathology. Thus, GVS could be potentially be used for vestibular rehabilitation and pre-habituation prior to surgical intervention.
Task Progress:	In the past year we have developed and validated a portable Galvanic Vestibular Stimulation (GVS) system that uses electrical stimulation of the vestibular system to model the effects of microgravity exposure. The system has been tested on 32 normal subjects and demonstrated that GVS produced postural and locomotor deficits observed following both short (shuttle) and long (ISS) duration missions. In addition, 7 veteran astronauts have participated in a demonstration of the technology, and all have reported that the subjective sensations of movement and the instability produced by the GVS system were remarkably similiar to that experienced after space flight.
Bibliography Type:	Description: (Last Updated: 09/07/2020)
Articles in Peer-reviewed Journals	MacDougall HG, Moore ST, Curthoys IS, Black FO. "Modeling postural instability with Galvanic vestibular stimulation." Exp Brain Res. 2006 Jun;172(2):208-20. Epub 2006 Jan 24. PMID: 16432695 , Jun-2006
Articles in Peer-reviewed Journals	Moore ST, MacDougall HG, Peters BT, Bloomberg JJ, Curthoys IS, Cohen, HS. "Modeling locomotor dysfunction following spaceflight with Galvanic vestibular stimulation." Exp Brain Res. 2006 Oct;174(4):647-59. Epub 2006 Jun 9. PMID: 16763834 , Oct-2006